DS-A3 | Regression Baseline and Diagnostics

1. Problem, data and scope

Question. Can bulk ∑REY (ppm) be predicted from non-REY trace and minor elements in these 29 samples, and where does the prediction stop being reliable?

Data source: Ali et al. (2026), *Journal of Asian Earth Sciences*.
DOI: 10.1016/j.jseaes.2025.106927.

2. Split and leakage audit

∑REY is exactly the sum of the 15 REY columns, and several ratio columns are computed from them, so none of those can be a predictor. The checks were run in code:

Check	Passed	Detail
∑REY equals the sum of La..Lu + Y in the file	yes	max abs difference 5.7e-14 ppm
Y/Ho equals Y divided by Ho	yes	max abs difference 0.0e+00
No predictor is a REY, derived ratio, major oxide, censored column or the target	yes	35 predictors
Predictors have no missing values	yes	
Predictors are strictly positive, so log10 is safe	yes	minimum value 0.005

Excluded group	Columns	Reason
Target	∑REY	the quantity being predicted
REY elements	La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Y, Ho, Er, Tm, Yb, Lu	the target is exactly their sum: direct leakage
Derived ratios	Ce/Ce*, Pr/Pr*, Y/Y*, Eu/Eu*, Y/Ho, La/Yb, La/Sm, Sr/Ca, Rb/Sr	computed from REY (leakage) or from other predictors (redundant)
Major oxides and LOI	17 columns	recorded for only 14 of 29 samples; using them would halve the data
Below-detection columns	Ag, Tl, Hf	contain '<' entries, which are not numbers

Frozen split. Stratified by layer, 2 test samples per layer, seed 42: 23 training and 6 test samples (test: KG-1, KG-15, KG-21, KG-22, KG-27, KG-7). It is not stratified by lithology, so the test set contains 0 chert-bearing samples and cannot show how the models behave on cherts. All 29 samples were profiled (missing values, skewness, correlations with ∑REY) before the split was frozen. No model was fitted or selected from that profile, but the test set is blind to model fitting, not to exploratory analysis.

3. Preprocessing, tuning and model comparison

Preprocessing, features and tuning. 35 non-REY trace and minor elements; log10 then standardised inside each cross-validation fold; ∑REY standardised inside the model and predicted back in ppm. Tuning used 5-fold x 10-repeat cross-validation on the 23 training samples only (RMSE in ppm): k in {1, 2, 3, 4, 5, 8, 12, 16, 22} for OLS on the k best predictors, and 40 penalty values for ridge (0.01 to 1000) and for lasso (0.001 to 1). Every tuning run is saved in results/tuning_runs.csv. The test set is not used for selection.

Model comparison. Errors are RMSE in ppm.

Model	Features used	CV RMSE, ppm (mean ± SD)	Train RMSE	Test RMSE, ppm [95% bootstrap CI]	Test gain vs naive
Naive: training mean	0	181.3 ± 46.7	182.2	169.4 [116, 211]	0%
Naive: mean per layer	0	182.9 ± 47.2	164.5	142.6 [86, 183]	16%
OLS, all 35 predictors	35	80.4 ± 30.7	0.0	101.4 [30, 168]	40%
OLS, k best (k = 8)	8	55.5 ± 19.2	34.2	86.6 [33, 137]	49%
Ridge	35	61.6 ± 15.4	37.5	42.8 [19, 62]	75%
Lasso (selected)	8	51.9 ± 17.1	38.5	82.1 [32, 118]	52%

Selection. The rule fixed before scoring (lowest mean CV RMSE) selects Lasso. Ridge has a lower test RMSE (43 ppm), but the test set has 6 samples and the intervals overlap, so the selection stands. OLS on all 35 predictors is a retained negative result: with 23 training samples (design-matrix rank 22) it interpolates the training data (train RMSE 0.0) and generalises worse than the penalised models.

Leave-one-layer-out check. RMSE in ppm on training samples only: train on two layers, predict the third.

Model	LPL held out	MPL held out	UPL held out
Naive: training mean	241	150	200
Naive: mean per layer	n/a	n/a	n/a
OLS, all 35 predictors	128	103	78
OLS, k best (k = 8)	97	93	93
Ridge	58	70	86
Lasso	102	49	76

4. Diagnostics and error cases (selected model)

Residuals and heteroscedasticity. Figure 1 shows the out-of-fold residuals. Breusch-Pagan p = 0.28; residual SD by prediction third is 36, 37 and 54 ppm (highest third 1.5 times the lowest): no clear heteroscedasticity (low power). Shapiro-Wilk on the residuals: p = 0.55. With 23 samples these tests have little power. Calibration: a calibration slope of 1.07 (95% interval 0.92 to 1.21).

Influence. Samples above the Cook's D alert level 4/n: KG-28 (Cook's D 0.53, leverage 0.61, leave-one-out error 15 ppm); KG-12 (Cook's D 0.22, leverage 0.49, leave-one-out error 39 ppm); KG-25 (Cook's D 0.22, leverage 0.26, leave-one-out error 128 ppm). The largest out-of-fold errors are KG-25 (+125 ppm), KG-26 (+107 ppm), KG-29 (-91 ppm).

Test-set error cases. Six samples, largest error first:

Sample	Layer	Lithology	Observed	Predicted	Error (obs − pred)
KG-1	LPL	Phosphorite	281	430	-149
KG-22	UPL	Phosphorite	362	474	-112
KG-7	LPL	Dolomatic Phosphorite	124	57	+67
KG-27	UPL	Phosphorite	540	511	+29
KG-15	MPL	Phosphorite	566	547	+19
KG-21	MPL	Phosphorite	549	544	+5

The two largest errors account for 86% of the squared test error, so one or two samples dominate the score.

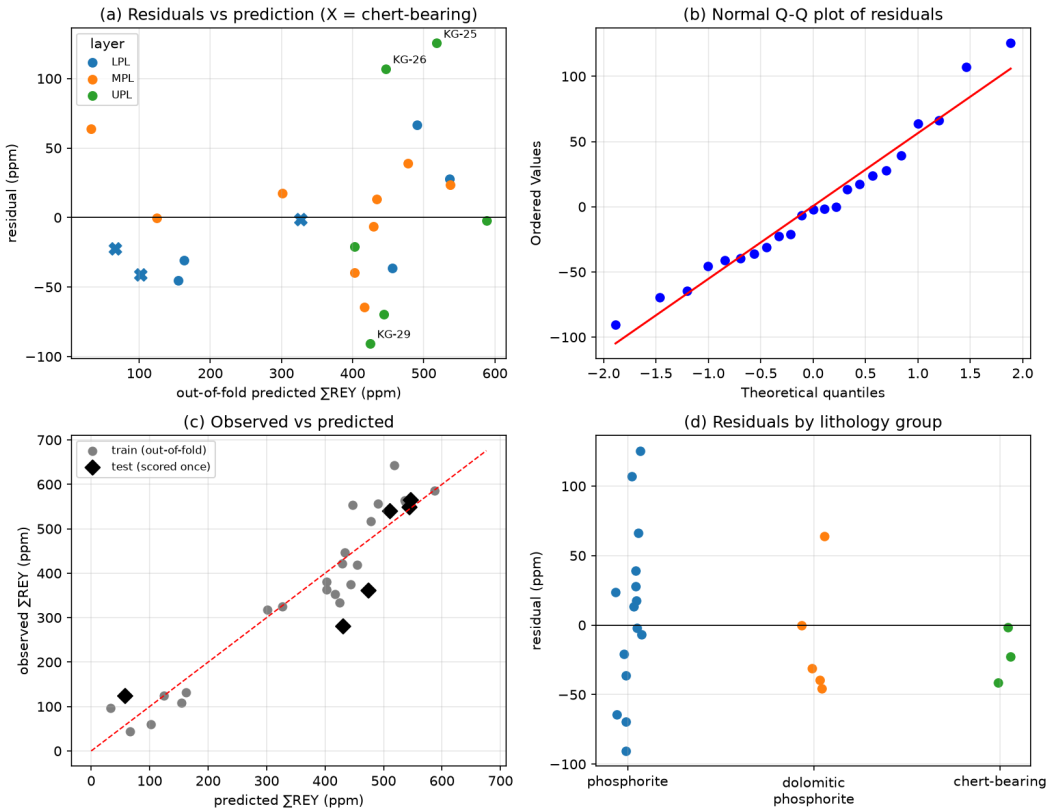


Figure 1. Out-of-fold residuals: (a) vs prediction, X = chert-bearing; (b) Q-Q plot; (c) observed vs predicted, training out-of-fold and test; (d) by lithology group.

Critical slices. Training samples, out-of-fold:

Slice type	Slice	n	RMSE, ppm	Mean error (obs − pred)	RMSE as % of slice mean
layer	LPL	8	38	-10	14%
layer	MPL	9	37	5	10%
layer	UPL	6	82	8	17%
lithology group	chert-bearing	3	27	-22	19%
lithology group	dolomitic phosphorite	5	42	-10	25%
lithology group	phosphorite	15	60	9	13%
observed level	low observed ∑REY	8	35	-8	23%
observed level	mid observed ∑REY	7	54	-47	14%
observed level	high observed ∑REY	8	66	50	12%

UPL has the largest absolute error (82 ppm); dolomitic phosphorite has the largest relative error (25%, n = 5). The 3 chert-bearing samples are over-predicted by 22 ppm on average and are all in the training set. Slices hold only 3 to 15 samples, so differences are suggestive at best. The mean error by observed level (-8, -47, 50 ppm) is the pattern that grouping on the observed value produces by itself; with the calibration slope above it is not read as bias.

Learning curve. Figure 2 shows the error on unseen training samples. Lasso RMSE falls from 110 ppm at 8 training samples to 64 ppm at 20, ridge from 89 to 67, while the naive mean stays near 189. The curves had not clearly flattened, so more samples might still help.

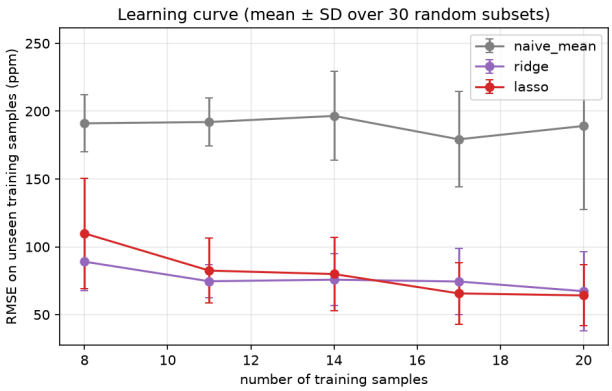


Figure 2. RMSE on unseen training samples against the number of training samples (30 random subsets each; hyper-parameters re-tuned; test set not used).

5. Complexity gain, and why the coefficients are not causal

Complexity gain. The gain from the naive mean to Lasso is large: CV RMSE 181 to 52 ppm, against a fold-to-fold SD of about 17 ppm. The further step from OLS on k best predictors to the selected model is 3.6 ppm, inside that spread, so the data do not clearly favour the more elaborate model. The mean per layer adds nothing (183 ppm): layer alone does not predict ∑REY here.

Coefficients. The five predictors most often selected by the lasso:

Predictor	Lasso coef (ppm per SD of log10)	Ridge coef (ppm per SD of log10)	Selected in 200 bootstrap fits
Be	89.1	26.8	99%
Ge	30.2	22.3	98%
U	0.5	18.8	61%
Sr	44.3	16.3	52%
Th	0.0	21.4	51%

- Correlated predictors share the signal, so the choice among them is unstable: Be is selected in 99% of bootstrap fits, but the next predictors in only about half, and lasso and ridge assign them different coefficients.
- Removing Be and re-tuning raises the lasso CV RMSE from 52 to 67 ppm (naive: 181); other correlated predictors recover much, not all, of the signal. Be correlates with Na % at r = 0.92.
- The data are observational. Apatite hosts REY together with elements such as Sr and U, so a shared mineral control is a more natural reading than an element-to-element effect. A coefficient describes this sample and is not the effect of changing an element.

6. Claim boundary

Supported. In these 29 samples, ∑REY can be predicted from a handful of trace elements with much lower error than the training mean, and the signal is shared by several correlated elements.

Not supported.

- Causation: no coefficient shows that an element controls REY enrichment.
- Generalisation: one section in one basin; nothing here supports other phosphorites, basins or ages.
- Fine model ranking: with 6 test and 23 training samples, differences of a few tens of ppm between candidates are within noise.
- Cherts: the test set has none and the training set has three.

- Independence: neighbouring samples in a section are similar, so the within-layer split and cross-validation are optimistic; leave-one-layer-out is more honest but also uncertain.
- Pre-registration: the analysis is retrospective, and the test set was not blind to the exploratory profiling.

7. Reproduction guide

- Create a virtual environment and install: *python -m venv .venv*, activate it, *python -m pip install -r requirements.txt* (exact versions in *requirements-lock.txt*).
- Save the original supplementary workbook as *data/raw/Supplementary file.xlsx* (not redistributed) and check that its SHA-256 equals: 17343594cd7317c950a5ab7fd713c0928859a0a333ac20798d9da6ce645dd18b
- Open *DS-A3_regression_baseline.ipynb* in VS Code, select the .venv kernel and run all cells from the top (about 4 minutes). Then run *python src/build_report.py* to regenerate this report.
- Everything is seeded (seed 42). Environment of this run: Python 3.13.9, numpy 2.5.4, pandas 3.1.0, scikit-learn 1.9.1, statsmodels 0.15.0; small differences across library versions are possible.
- AI assistance (Claude, Anthropic) is disclosed in the README and in *docs/AI_use_log.md*.